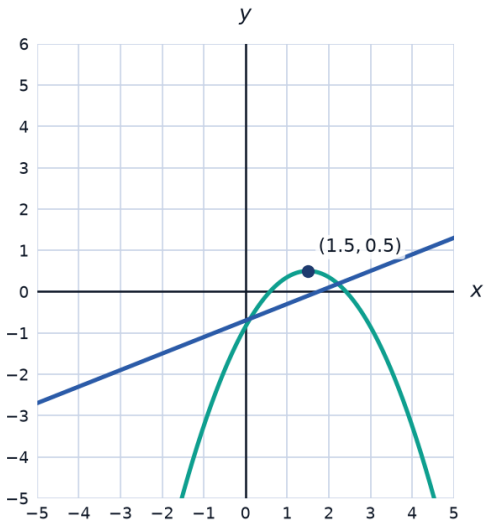
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Permutations**

How many ways can 5 distinct books be arranged on a shelf?

2 **[Foundational] • Target: Permutations**

In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 12 members?



3 **[Foundational] • Target: Permutations**

How many distinct arrangements are there of the letters in the word APPLE?

4 **[Foundational] • Target: Combinations**

How many ways can a committee of 4 be chosen from 11 people?

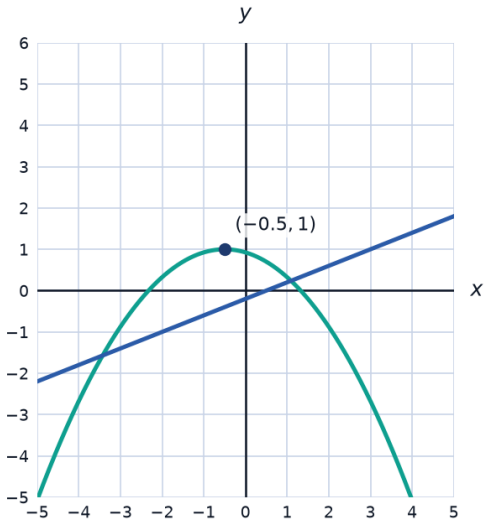
5 **[Foundational] • Target: Combinations**

A pizza shop offers 8 toppings. How many different 3-topping pizzas can be made?

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6 [Foundational] • Target: Combinations

From a group of 6 men and 5 women, how many ways can a committee of 4 be formed with exactly 2 men and 2 women?



7 [Foundational] • Target: Probability rules

A fair die is rolled. Find $P(\text{even or greater than } 4)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Probability rules

$P(A) = 0.45$ and $P(B) = 0.3$, and A and B are mutually exclusive. Find $P(A \text{ or } B)$.

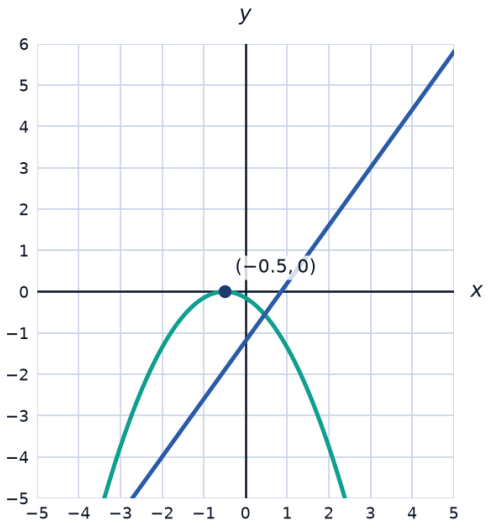
9 [Intermediate] • Target: Probability rules

A card is drawn from a standard 52-card deck. Find $P(\text{not a face card})$.

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10 [Intermediate] • Target: Conditional probability and independence

$P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$. Are A and B independent?



11 [Intermediate] • Target: Conditional probability and independence

In a class, 60% of students play sports, and 25% of all students both play sports and are honor students. Find $P(\text{honor student} \mid \text{plays sports})$.

12 [Intermediate] • Target: Conditional probability and independence

Events C and D are independent, with $P(C) = 0.7$ and $P(D) = 0.6$. Find $P(C \cup D)$.

13 [Intermediate] • Target: Measures of center and spread

Find the mean and median of the data set $12, 15, 18, 20, 25$.

14 [Intermediate] • Target: Measures of center and spread

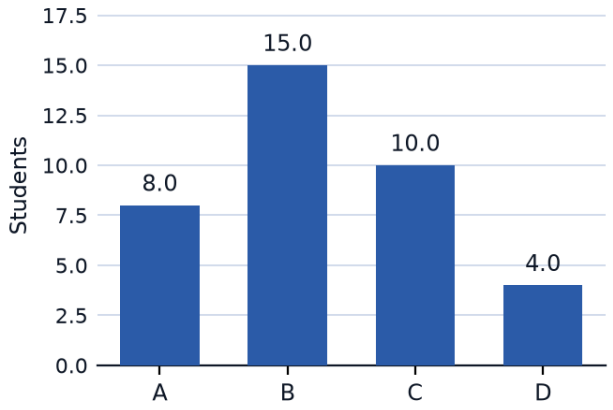
Find the population standard deviation of the data set $4, 8, 6, 5, 3$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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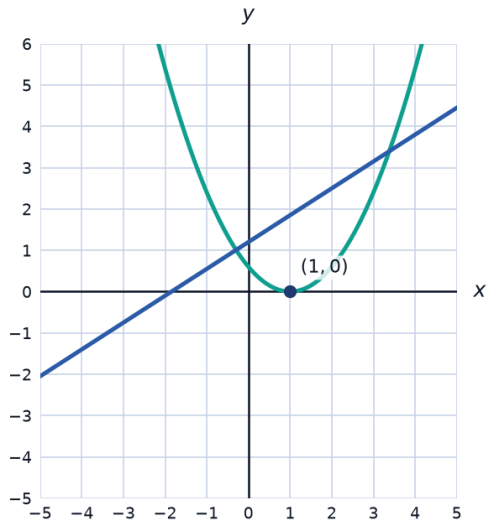
15 [Challenging] • Target: Data representation • Cross-Lesson: 1.01 Advanced Algebra

The bar chart shows the distribution of grades in a class of 37 students. Find the probability that a randomly selected student earned a grade of A or B, as a percent to the nearest whole percent.



16 [Challenging] • Target: Data representation • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Using the same bar chart, find the probability that a randomly selected student scored a C or D grade, as a percent to the nearest whole percent.



17 [Challenging] • Target: The binomial theorem • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Find the coefficient of x^3y^2 in the expansion of $(x + y)^5$.

18 [Challenging] • Target: The binomial theorem • Cross-Lesson: 4.01 Lines, Triangles & Polygons

In the expansion of $(2x - 1)^4$, find the coefficient of x^2.

19 [Challenging] • Target: The normal distribution • Cross-Lesson: 5.01 Trigonometry

A data set is normally distributed with mean 100 and standard deviation 15. What percent of values lie between 70 and 130?

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20 [Challenging] • Target: The normal distribution • Cross-Lesson: 6.01 Circles & Coordinate Geometry

For the same distribution (mean 100, standard deviation 15), find the approximate percentage of values greater than 115.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Permutations

Find the population variance of the data set: $2, 8, 12, 8, 10$.

a. 173/15

b. 56/5

c. 677/60

d. 163/15

22 [Foundational] • Target: Combinations

Find the median of the data set: $9, 4, 9, 2, 7$.

a. 17

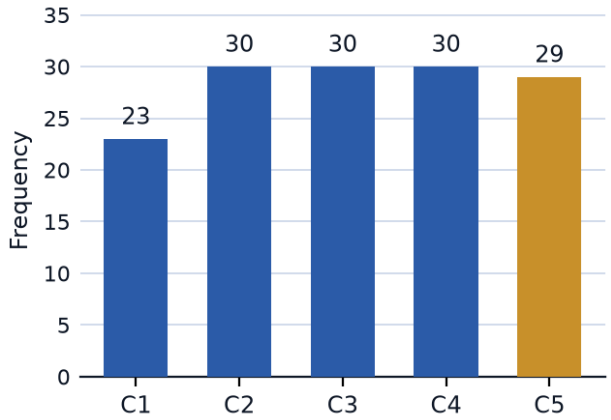
b. 24

c. 13

d. 7

23 [Foundational] • Target: Probability rules

Find the mean of the data set: $23, 11, 31, 33, 30, 34$.



a. 25

b. 27

c. 5

d. 42

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24 [Foundational] • Target: Conditional probability and independence

A data set: 2, 5, 5, 7, 9. The mode is:

a. 5

b. 6

c. 7

d. 2

25 [Foundational] • Target: Measures of center and spread

68% of data in a normal distribution falls within $0.5cm$ standard deviations of the mean.

a. 3

b. 2

c. 1

d. 1.5

26 [Foundational] • Target: Data representation

95% of data in a normal distribution falls within $0.5cm$ standard deviations.

a. 1

b. 3

c. 2

d. 1.5

27 [Foundational] • Target: The binomial theorem

The bar chart below shows the number of students who chose different sports as their favorite. What is the total number of students surveyed?

a. 50

b. 60

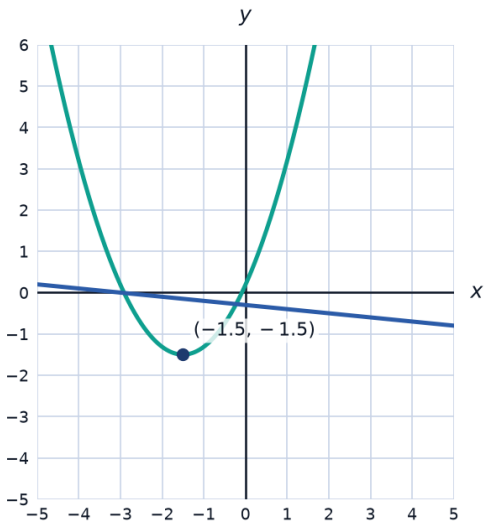
c. 70

d. 80

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28 [Foundational] • Target: The normal distribution

The following bar chart shows the number of students who prefer different fruits. What is the total number of students who prefer apples and bananas?



a. 20

b. 25

c. 30

d. 35

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: quality control sampling

The bar chart below shows the number of students who achieved different scores on a recent quiz. What is the total number of students who scored 8 or 10?

a. 25

b. 30

c. 35

d. 40

30 [Intermediate] • Target: Permutations

The table shows exam marks for 20 students. Find the mean and identify the modal class.

a. Mean ≈ 65, Modal class 60–69

b. Mean ≈ 68, Modal class 70–79

c. Mean ≈ 60, Modal class 50–59

d. Mean ≈ 72, Modal class 60–69

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31 [Intermediate] • Target: Combinations

In a group of 20 students, 9 study French, and 1 study both French and Spanish. If a student is chosen at random from those studying French, what is the probability they also study Spanish?

a. 1/9

b. 7/36

c. 19/90

d. 5/18

32 [Intermediate] • Target: Probability rules

A bag has 23 marbles, of which 2 are red. What is the probability of drawing a red marble?

a. 43/230

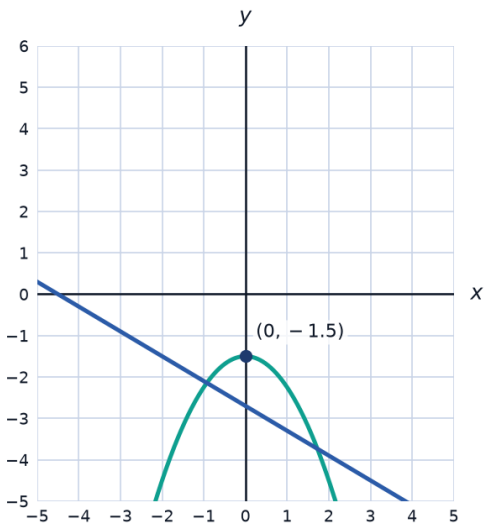
b. 2/23

c. 89/230

d. 33/115

33 [Intermediate] • Target: Conditional probability and independence

In how many ways can 2 of 7 books on a shelf be arranged?



a. 82

b. 33

c. 42

d. 341

34 [Intermediate] • Target: Measures of center and spread

In how many ways can 3 of 4 photos in an album be arranged?

a. 132

b. 69

c. 154

d. 24

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35 [Intermediate] • Target: Data representation

In how many ways can 2 of 4 runners in a race (1st through r-th place) be arranged?

a. 296

b. 112

c. 12

d. 270

36 [Intermediate] • Target: The binomial theorem

A basketball player makes a free throw with probability $2/3$. If she takes 2 free throws, what is the probability she makes at least one?

a. 53/90

b. 55/72

c. 8/9

d. 35/36

37 [Intermediate] • Target: The normal distribution

A committee of 5 people is chosen from 9 people, and then 4 of those 5 are assigned distinct roles (e.g., chair, secretary, ...). How many ways can this be done?

a. 2696

b. 15120

c. 396

d. 4484

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: quality control sampling • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Events A and B are mutually exclusive, with $P(A) = 2/7$ and $P(B) = 3/8$. Find $P(A \text{ or } B)$.

a. 37/56

b. 269/280

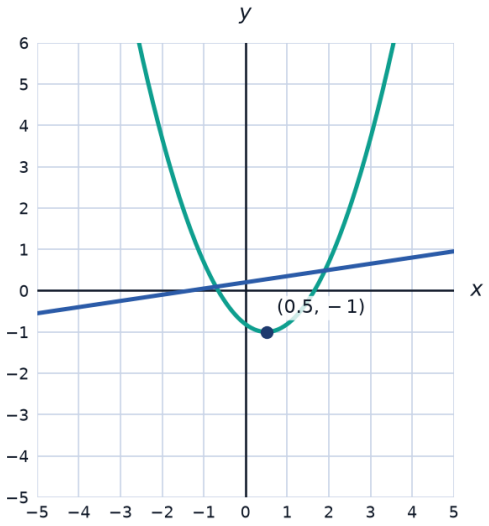
c. 55/168

d. 83/168

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39 [Challenging] • Target: Permutations • Cross-Lesson: 5.01 Trigonometry

The scatter plot shows the relationship between temperature (°C) and ice-cream sales (hundreds). Estimate the sales when temperature is 30°C.



a. About 300

b. About 400

c. About 500

d. About 600

40 [Challenging] • Target: Combinations • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Which measure is most affected by an outlier?

a. Median

b. Mode

c. Mean

d. Range (equally)

41 [Challenging] • Target: Probability rules • Cross-Lesson: 7.01 Solid Geometry & Similarity

A team of 4 is chosen from 9 players. How many different teams are possible?

a. 120

b. 126

c. 3024

d. 84

42 [Challenging] • Target: Conditional probability and independence • Cross-Lesson: 1.01 Advanced Algebra

A bag has 10 marbles, of which 4 are blue. Two marbles are drawn without replacement. What is the probability both are blue?

a. 1/30

b. 1/20

c. 2/15

d. 7/30

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43 [Challenging] • Target: Measures of center and spread • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

In how many ways can 3 of 9 students in a row be arranged?

a. 162

b. 531

c. 352

d. 504

44 [Challenging] • Target: Data representation • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

A committee of 6 people is chosen from a group of 9 people. How many different committees are possible?

a. 84

b. 142

c. 111

d. 134

45 [Challenging] • Target: The binomial theorem • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Show that the binomial distribution approaches Poisson as $n \to \infty$, $p \to 0$, $np = \lambda$ fixed.

a. Requires $n > 100$

b. Not true

c. $\binom{n}{k} p^k (1-p)^{n-k} \to \frac{e^{-\lambda}\lambda^k}{k!}$ — use $(1-\lambda/n)^n \to e^{-\lambda}$

d. Only for $k = 0$

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